Data Analytics Technology

Databases for Data Analytics

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Outline

- Introduction
- Operational vs Analytical Databases
- OLTP vs OLAP Differences and Use Cases
- Entity-Relationship Diagrams
- Indexing
- SQL
- Postgres hands-on example

Introduction

- Motivation for Using Databases
- Key advantages: Organization, Efficiency, Integrity, Scalability
- Example: Using databases vs. spreadsheets
- Share experiences and knowledge

Operational vs. Analytical Databases (revisited)

- Operational Databases (OLTP): High volume, quick transactions
- Analytical Data Warehouses (OLAP): Complex queries, historical analysis

- Comparison: Key differences and complementary use
- Examples: Banking systems (OLTP) vs. sales dashboards (OLAP)

OLTP vs. OLAP: Differences and Use Cases (revisited)

- Performance Metrics: Speed, transaction volume, data structure
- Examples: Inventory management (OLTP) vs. sales reports (OLAP)
- Why Both Are Important: Complementary roles

- Which type for an e-commerce website?

Entity-Relationship (ER) Diagrams

- Basic Concepts: Entities, Attributes, Relationships
- Primary and Foreign Keys
- Example: Simple ER diagram

Entities, Attributes

- Entities:
  - Entities represent real-world objects or concepts that have a distinct existence. For example, in a university database, entities might be Student, Course, Professor, etc.
  - Entities are typically represented as tables in a database.
  - Each entity has attributes that describe its properties. For example, a Student entity might have attributes like StudentID, Name, Email, etc.

- Attributes:
  - Attributes are the data elements that describe the characteristics of an entity.
  - For example, Name and Email are attributes of the Student entity.
  - Attributes can be of different data types (string, integer, date, etc.) and may have constraints (e.g., NOT NULL).

Relationships

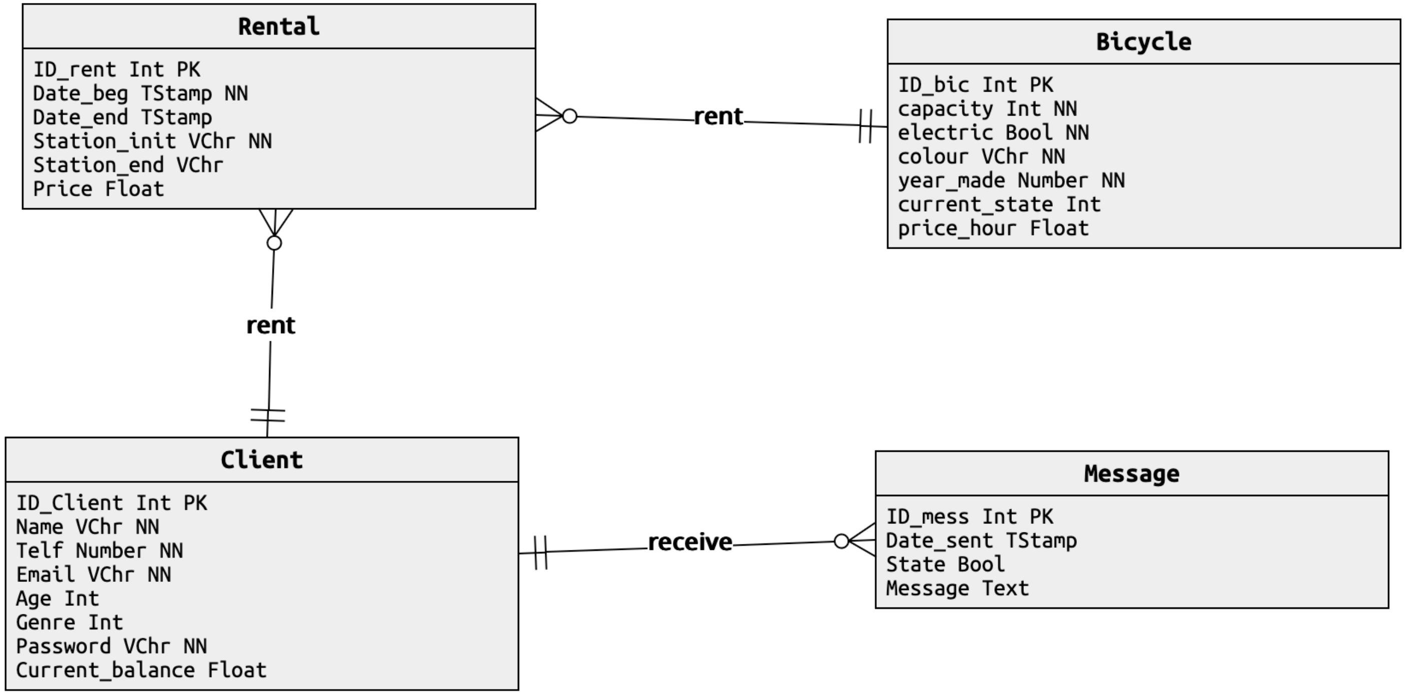
- Relationships define how entities are connected to each other.
- For instance, a Student may Enroll in a Course. Here, Enroll is a relationship that links Student and Course.
- Relationships can have attributes of their own. For example, the Enroll relationship might have an attribute called DateOfEnrollment.
- They can be of different types: One-to-One, One-to-Many, and Many-to-Many.

Primary and Foreign Keys

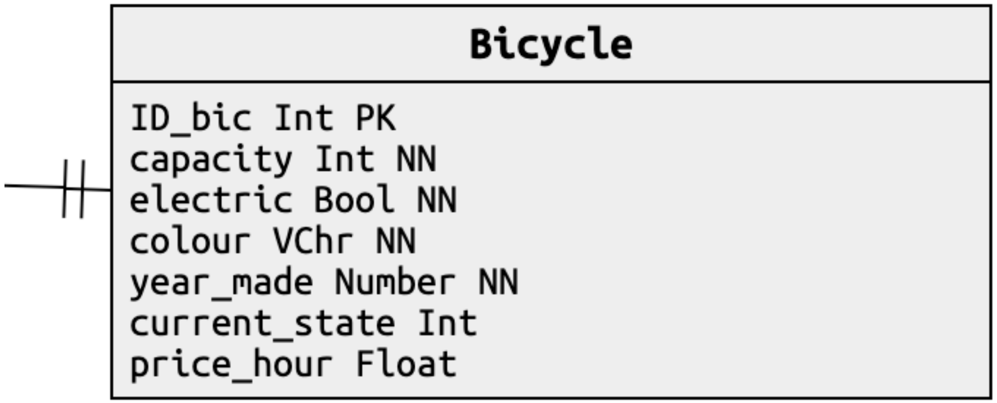
- Primary Key (PK):
  - A primary key is a unique identifier for each record in an entity (table). It ensures that each entry is uniquely identified.
  - Example: In a Student table, StudentID could be a primary key because it uniquely identifies each student.
  - Primary keys cannot have NULL values, and they must contain unique values.

- Foreign Key (FK):
  - A foreign key is an attribute in one table that links to the primary key of another table, creating a relationship between the two tables.
  - Example: If StudentID is a primary key in the Student table, it could be used as a foreign key in the Enrollment table to show which student is enrolled in which course.
  - Foreign keys help maintain referential integrity, ensuring that the relationships between tables are preserved.

Example of an ER Diagram



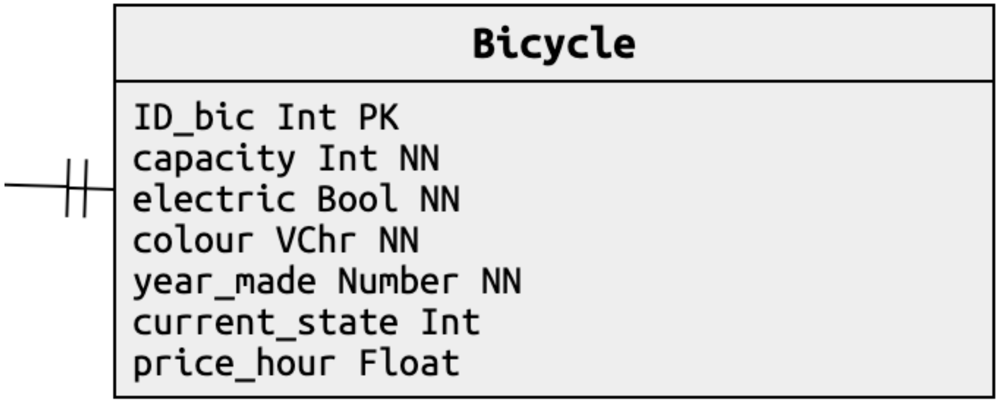
Example of an ER Diagram




```
CREATE TABLE bicycle (
        id_bic        INTEGER,
        capacity      INTEGER NOT NULL,
        electric      BOOL NOT NULL,
        colour        VARCHAR(512) NOT NULL,
        year_made               NUMERIC(4,0) NOT NULL,
        current_state INTEGER,
        price_hour              FLOAT(8),
        PRIMARY KEY(id_bic)
);

ALTER TABLE bicycle ADD CONSTRAINT constraint_0
CHECK (year_made > 2000);
```

Example of an ER Diagram

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        colour        VARCHAR(512) NOT NULL,
        year_made     NUMERIC(4,0) NOT NULL,
        current_state INTEGER,
        price_hour    FLOAT(8),
        PRIMARY KEY(id_bic)
);
```

```
insert into bicycle
values(1,1,FALSE,'blue',2020,0,10);
```

Indexing

- An index is a data structure that improves the speed of data retrieval operations on a database table. Think of it like the index in a book, which helps you quickly locate a topic.
- Purpose:
  - Speeds Up Queries: Allows for faster searches, filtering, and sorting.
  - Reduces I/O Operations: Helps the database locate data without scanning the entire table.
- How It Works:
  - An index is created on one or more columns of a table.
  - It stores the values of these columns and pointers to the actual data rows.
- Example:
  - Searching for EmployeeID = 12345 without an index might mean scanning every record in the Employee table.
  - With an index on EmployeeID, the database can directly jump to the location of 12345.

Types of Indexes

- **Primary Index**: Automatically created on the primary key. Ensures that every row is uniquely identifiable.
- **Secondary Index**: Manually created on non-primary key columns to improve search performance. E.g., indexing LastName in an Employee table for faster name searches.
- **Composite Index**: An index on multiple columns. Useful for queries that filter or sort on more than one column.
- **Unique Index**: Ensures that all values in the indexed column are unique, preventing duplicates.

- **Pros**: Faster queries, better performance on searches, filtering, and sorting.
- **Cons**: Slower INSERT, UPDATE, and DELETE operations due to maintaining the index. Additional storage space required.
- Use Indexes Wisely: Only create indexes on columns that are frequently searched, filtered, or used in JOIN operations.

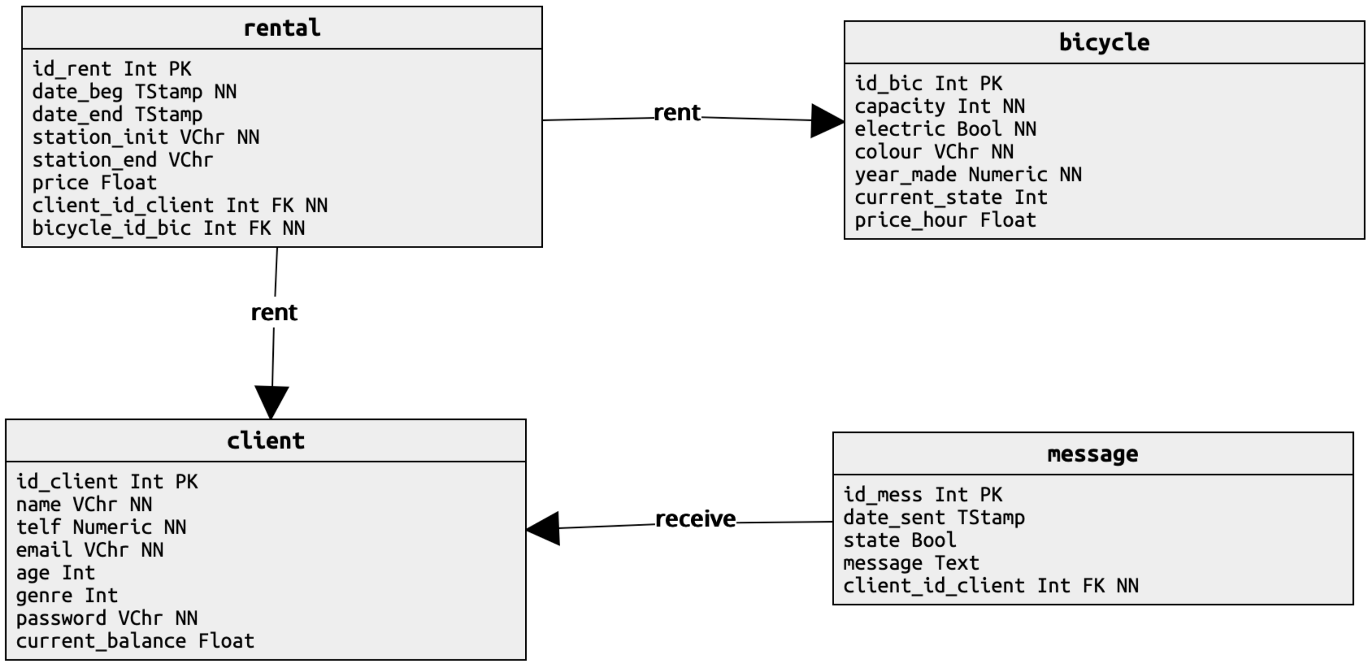
Introduction to SQL and PostgreSQL

- SQL (Structured Query Language) - Language for interacting with databases

- Basic SQL Commands: SELECT, INSERT, UPDATE, DELETE
- More commands: JOIN, COUNT, SUM, AVG, GROUP BY, LIMIT, ORDER BY

- Why PostgreSQL? Open-source, extensible, supports complex queries
- Interactive Demo: Setting up PostgreSQL, connecting, running queries

Queries



- List all bicycles?
- What is the name of client with id_client 1?
- How many byclicles are there in the DB?
- How many byclicles are there in the DB from 2023?
- How many rentals does Clara have?

Queries

- List all bicycles?
- What is the name of client with id_client 1?
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Bicycle rental Decision Support System (revisited)

The administrators of the bike sharing company intend to obtain strategic information for decision support from the data in its operational database for instance for determining the most solicited bike types or the busiest stations. For this they decided to order a Data Warehousing and OLAP solution. This solution should enable the following four (4) questions to be answered or the requested information be presented, while at the same time promoting the ad-hoc exploitation of the data for the discovery of "new knowledge":

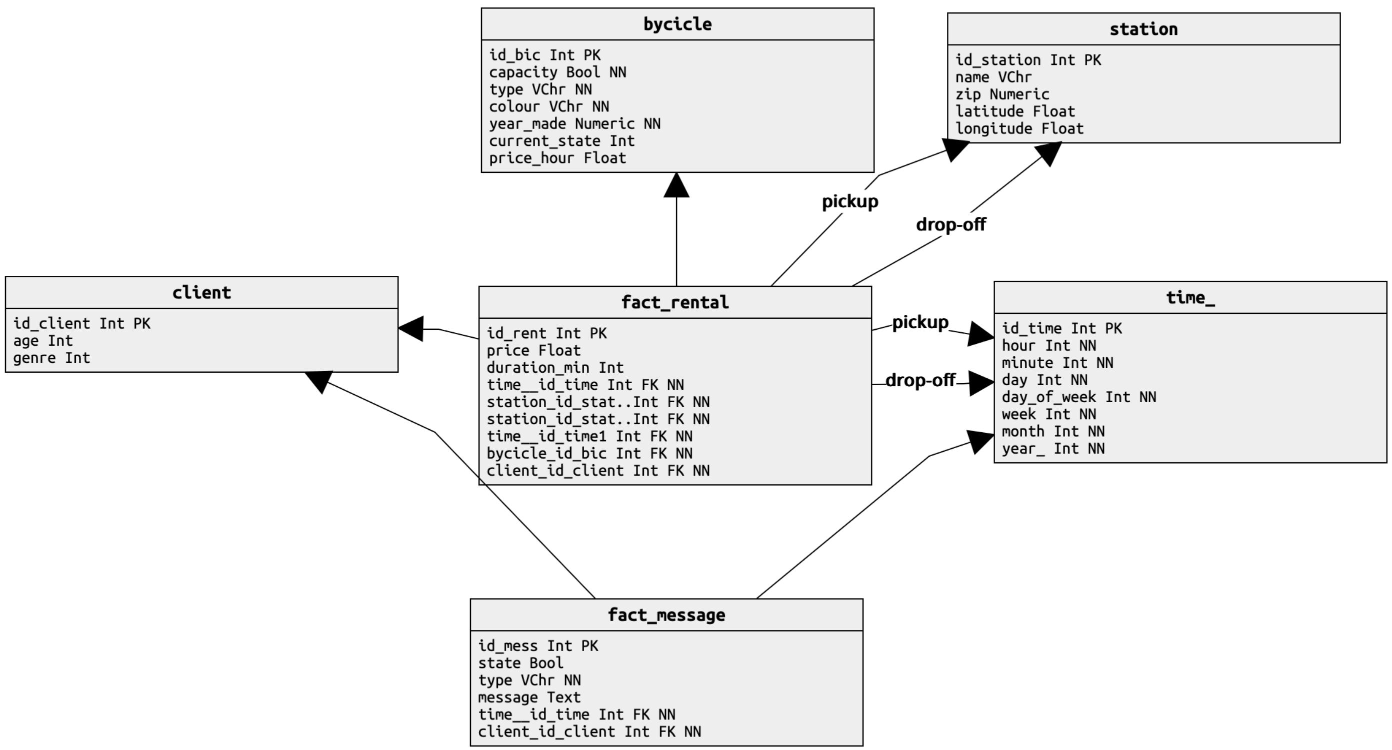
1. Which are the busiest stations in terms of number of rentals and revenue? How does this evolve over the day/week/year?

2. Is there a significant difference between men and women clients, namely in terms of rental time, bike types and rental duration?

3. Does age play a role in the rentals, namely regarding price and rental duration?

4. Is there a relation between the reading of promotion messages (type=general) by the clients and their rentals?

Proposal of datawarehouse (revisited)



1. Which are the busiest stations in terms of number of rentals and revenue? How does this evolve over the day/week/year?

```
SELECT  t.month, t.day_of_week, s.name ”station”,
             count(*) as “number of rentals”, sum(r.price) as “revenue”

FROM rentals r, time t, station s
WHERE <JOIN_CLAUSES>
AND t.year = 2024

GROUP BY s.name, t.month, t.day_of_week

ORDER BY 4,5;
```

2. Is there a significant difference between men and women clients, namely in terms of rental time, bike types and rental duration?

3. Does age play a role in the rentals, namely regarding price and rental duration?

4. Is there a relation between the reading of promotion messages (type=general) by the clients and their rentals?

Dataedo /cartoon

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